

LexMachina at SemEval-2026 Task 2: Predicting Variation in Emotional Valence and Arousal over Time from Ecological Essays

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Abstract

Tracking emotional dynamics like valence and arousal is critical for understanding user baselines in ecological text. However, encoder models often struggle to distinguish stable user traits from dynamic shifts, leading to poor generalization. We present LexMachina for SemEval-2026 Task 2, addressing “domain shift” and “regression to the mean.” Our system utilizes a DeBERTa-v3-Base backbone with a bifurcated strategy: post-hoc Isotonic Regression for valence calibration and a Domain Adversarial Neural Network (DANN) to mitigate user-bias in arousal. LexMachina achieved composite scores of $r = 0.645$ (Valence) and $r = 0.434$ (Arousal), demonstrating that adversarial disentanglement effectively captures nuances in longitudinal affective data.

Keywords: Longitudinal Affect Assessment, Valence-Arousal, DeBERTa, Domain Adversarial Neural Networks (DANN), Isotonic Regression, Gradient Reversal.

1 Introduction

Affective quantification traditionally relies on the Valence-Arousal space (Russell, 1980). In this work, we address SemEval-2026 Task 2 (Subtask 1) by predicting continuous valence-arousal trajectories from longitudinal essays. A primary challenge in this domain is annotator idiosyncrasy and label noise. We propose a multi-stage pipeline built on DeBERTa-v3-base embeddings. To ensure signal integrity, we implemented an Out-of-Fold (OOF) Data Sanitation Protocol, purging 9%

of the training data exhibiting extreme deviation ($\delta > 1.5$) from model consensus. Our modeling approach is bifurcated: Valence is refined via post-hoc Isotonic Regression to fix calibration drift, while Arousal is modeled through a Domain Adversarial Neural Network (DANN) to disentangle user-specific biases from universal emotional signals.

2 Background and Related Work

2.1 Related Work

Transformer-based architectures have set benchmarks in mapping text to continuous V-A space (Azevedo Mendes and Martins, 2023). Recent innovations include integrating VAD features into attention mechanisms (Moreno Novoa et al., 2025) and utilizing ordinal classification to minimize perceptual errors (Mitsios et al., 2024). For low-resource or domain-specific settings, strategies like RAG and multi-teacher pseudo-labeling have proven effective for cross-domain adaptation (Jian et al., 2025). Our work builds on these by applying adversarial disentanglement to longitudinal user data.

2.2 Task Definition and Evaluation

The task requires mapping a user’s chronological stream $S = \{e_1, e_2, \dots, e_m\}$ to bivariate outputs (v_i, a_i) where $v \in [-2, 2]$ and $a \in [0, 2]$. Evaluation relies on a Composite Correlation (r_{comp}), aggregating inter-user traits ($r_{between}$) and temporal fluctuations (r_{within}) via the Fisher-Z transfor-

Algorithm 1 Adversarial Training Loop for Arousal Stream

Require: Sanitized Corpus $\mathcal{S}_{clean}$, Max Epochs E , Total Steps T **Require:** Feature Extractor G_f , Regressor G_y , Discriminator G_d

```
1: Initialize network parameters  $\theta_f, \theta_y, \theta_d$ 
2:  $t \leftarrow 0$  ▷ Current step counter
3: for  $epoch = 1$  to  $E$  do
4:   for each batch  $(X, Y_{arousal}, U_{id})$  in  $\mathcal{S}_{clean}$  do
5:      $p \leftarrow \frac{t}{T}, \lambda \leftarrow \frac{2}{1+\exp(-10 \cdot p)} - 1$  ▷ Dynamic GRL schedule
6:      $H \leftarrow G_f(X; \theta_f), \hat{Y} \leftarrow G_y(H; \theta_y), \hat{U} \leftarrow G_d(H; \theta_d)$  ▷ Forward Pass
7:      $\mathcal{L}_{task} \leftarrow \alpha \text{MSE}(Y, \hat{Y}) + (1 - \alpha) \text{CCC}(Y, \hat{Y})$ 
8:      $\mathcal{L}_{domain} \leftarrow \text{CrossEntropy}(U, \hat{U})$ 
9:      $\nabla_{\theta_f} \leftarrow \frac{\partial \mathcal{L}_{task}}{\partial \theta_f} - \lambda \frac{\partial \mathcal{L}_{domain}}{\partial \theta_f}$  ▷ Backward Pass (with GRL)
10:    Update parameters  $\theta_y, \theta_d, \theta_f$  and increment  $t \leftarrow t + 1$ 
11:  end for
12: end for
```

66 mation:

$$r_{comp} = \tanh \left(\frac{1}{2} \left(\text{arctanh}(r_{between}) + \text{arctanh}(r_{within}) \right) \right) \quad (1)$$

2.3 Dataset Description

68 The dataset consists of longitudinal essays and
69 feeling words. Our partition strategy highlights
70 the stylistic gap addressed by our adversarial ap-
71 proach (Table 1).

Table 1: Dataset Statistics.

Partition	User Type	# Users	# Texts	Val ($\mu \pm \sigma$)	Aro ($\mu \pm \sigma$)
Train	Seen	137	2,487	0.22 ± 1.29	0.75 ± 0.76
Dev	Seen	92	277	0.22 ± 1.30	0.75 ± 0.74
Test	Unseen	91	1,737	Hidden	Hidden

3 System Overview

73 We propose a dual-stream architecture designed to
74 address the distinct generalization challenges of
75 the “Seen” versus “Unseen” user cohorts. While
76 both streams utilize a shared linguistic backbone,
77 they employ divergent optimization strategies: a
78 direct regression approach with post-hoc cali-
79 bration for Valence, and a Domain-Adversarial
80 framework for Arousal.

3.1 Data Sanitation Protocol

82 Preliminary error analysis revealed that the train-
83 ing corpus contained a subset of samples with

“noisy” labels—specifically, instances where the
ground truth annotation sharply contradicted the
semantic polarity of the text. To prevent the model
from learning incoherent decision boundaries, we
implemented an automated sanitation protocol.

We performed a 3-Fold Out-of-Fold (OOF)
Cross-Validation on the training set using a pre-
liminary DeBERTa regressor. Samples where
the absolute error between the prediction and the
ground truth exceeded a strict threshold ($\delta >$
1.5) were identified as “poisonous” and discarded.
This process removed approximately 9% of the
data, resulting in a sanitized corpus ($\mathcal{S}_{clean}$) that
provided a more consistent training signal.

3.2 Linguistic Backbone

For feature extraction, we utilize **DeBERTa-v3-
base** (He et al., 2020). While Transformer-based
models generally outperform recurrent architec-
tures due to bidirectional attention, DeBERTa was
selected specifically for its *disentangled attention
mechanism*. This allows the model to separate
syntactic structure from semantic content, which
is critical for parsing the informal, “telegraphic”
nature of social media text observed in this task.

We extract the contextualized word embeddings
from the final hidden layer and aggregate them us-
ing Max-Pooling (alongside the [CLS] token) to
capture the most salient features of the input se-
quence x_i :

$$h_i = \text{MaxPool}(\text{DeBERTa}(x_i)) \quad (2)$$

where $h_i \in \mathbb{R}^{768}$ serves as the latent affective rep-
resentation.

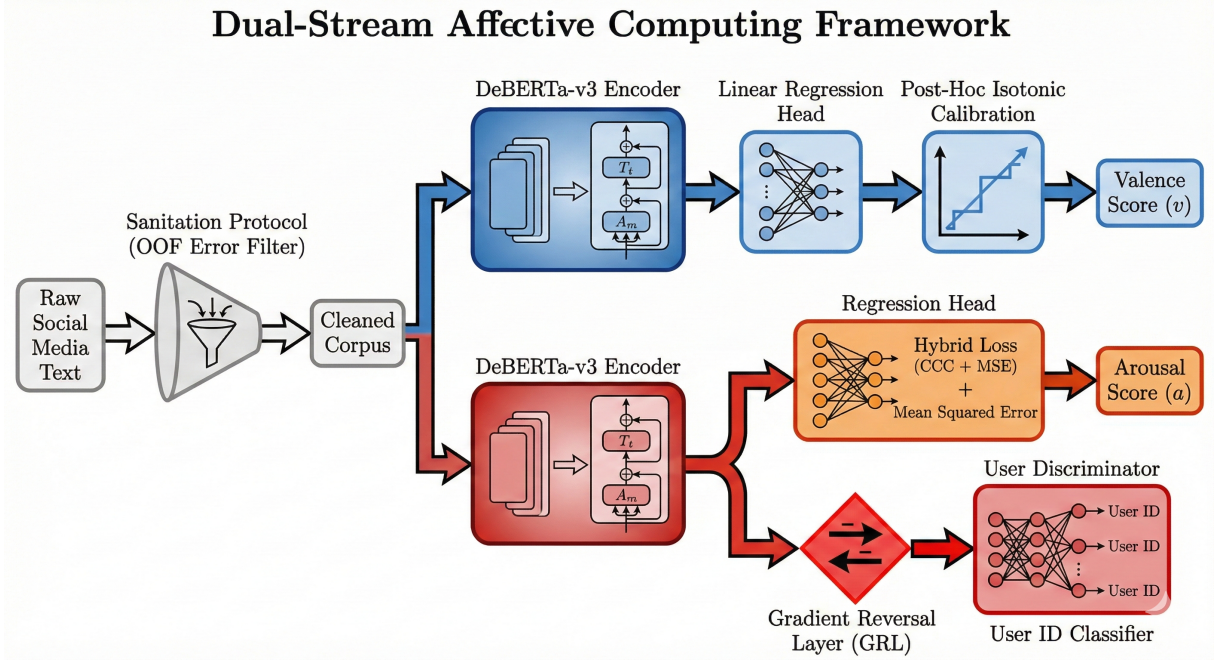


Figure 1: **System Architecture.** The pipeline splits into two parallel streams after the DeBERTa-v3 backbone. The Valence stream (top) utilizes a standard regression head followed by Isotonic Calibration to map predictions to the target distribution. The Arousal stream (bottom) employs a Domain-Adversarial Neural Network (DANN) with a Gradient Reversal Layer (GRL) to enforce user-invariant feature learning, mitigating the generalization gap for unseen subjects.

3.3 Valence Stream: Regression & Calibration

The Valence stream projects the latent vector h_i through a standard regression head followed by a linear layer. During development, we observed that the model exhibited a “conservative” bias, tending to predict values clustered around the mean ($v \approx 0$) rather than the extremes.

To address this, we applied **Post-Hoc Isotonic Calibration** (Zadrozny and Elkan, 2002). This non-parametric technique learns a monotonic function f_{iso} that maps the raw predictions to the broader cumulative distribution function (CDF) of the validation set:

$$\hat{v}_{final} = f_{iso}(\hat{v}_{raw}) \quad (3)$$

3.4 Arousal Stream: Domain-Adversarial Learning (DANN)

For Arousal, we identified a critical “User Generalization Gap.” Statistical analysis of the test partition revealed that while unseen users comprised 49.5% of the subject pool, they accounted for **60.2% of the total text volume** (1,045 out of 1,737 texts). This heavy skew towards unseen

subjects meant that standard supervision models, which often overfit to the writing styles of “Seen” users, would fail to generalize.

To mitigate this, we implemented a **Domain-Adversarial Neural Network (DANN)** (Ganin et al., 2016). The architecture consists of three components:

- **Feature Extractor (G_f):** The shared DeBERTa backbone.
- **Label Predictor (G_y):** A regressor predicting arousal intensity using a Hybrid Loss (CCC + MSE).
- **User Discriminator (G_d):** An adversarial classifier attempting to predict the User ID (u_i) from h_i .

We introduce a **Gradient Reversal Layer (GRL)** between the feature extractor and the discriminator. During backpropagation, the GRL reverses the gradients flowing from the discriminator:

$$\nabla \mathcal{L}_{adv} \rightarrow -\lambda \nabla \mathcal{L}_{adv} \quad (4)$$

This creates a min-max game: the discriminator tries to identify the user, while the feature extrac-

tor tries to “fool” it by learning *user-invariant* representations.

4 Experimental Setup

4.1 Implementation Details

All models were implemented using PyTorch and the HuggingFace transformers library, and trained on a single NVIDIA Tesla T4 GPU. The textual inputs were tokenized using the microsoft/deberta-v3-base tokenizer, with sequences truncated or padded to a maximum length of 128 tokens.

Both streams utilized the AdamW optimizer with a learning rate of 2×10^{-5} . To prevent overfitting, the Arousal DANN was trained for 4 epochs, whereas the simpler Valence regressor was trained for 3 epochs. A training batch size of 16 and an inference batch size of 32 were maintained across all experiments.

4.2 Ensemble Strategy

To ensure robust generalization, our final submission utilized a 3-seed ensemble strategy (Seeds = 42, 43, 44). For each seed, models were trained from scratch. The final predicted values represent the arithmetic mean of the predictions across all three independent seed runs.

5 Results and Analysis

5.1 Official Results and Cohort Analysis

Table 2 presents the performance of the LexMachina architecture across the official evaluation criteria. Our system achieved an overall composite correlation (r_c) of **0.645** for Valence and **0.434** for Arousal.

For **Valence**, performance remained remarkably stable across both cohorts ($r_c = 0.636$ for Seen, 0.669 for Unseen). This consistency validates our use of DeBERTa paired with post-hoc Isotonic Calibration, which successfully mapped texts to the continuous $[-2, 2]$ space without overfitting to specific user histories.

For **Arousal**, the cohort breakdown reveals a striking validation of our Domain-Adversarial Neural Network (DANN). Designed to explicitly optimize for zero-shot generalization by stripping

Table 2: **Official Subtask 1 Results.** Performance broken down by user cohort and input modality. Metrics are Pearson correlations: Composite (r_c), Between-User (r_b), and Within-User (r_w).

Evaluation Slice	Valence			Arousal		
	r_c	r_b	r_w	r_c	r_b	r_w
Overall System	0.645	0.712	0.567	0.434	0.461	0.406
<i>User Generalization Breakdown</i>						
Seen Users	0.636	0.718	0.537	0.343	0.323	0.363
Unseen Users	0.669	0.734	0.593	0.574	0.681	0.443
<i>Modality Breakdown</i>						
Words Only	0.655	0.730	0.563	0.572	0.631	0.507
Essay Only	0.627	0.665	0.586	0.307	0.315	0.298

user identity markers, the DANN yielded an exceptional Unseen Arousal score of $r_c = 0.574$. However, this adversarial disentanglement introduced a stark generalization trade-off. By penalizing the encoding of user-specific traits, the model ignored the highly predictive historical baselines of known individuals, causing the Seen Arousal score to drop to $r_c = 0.343$.

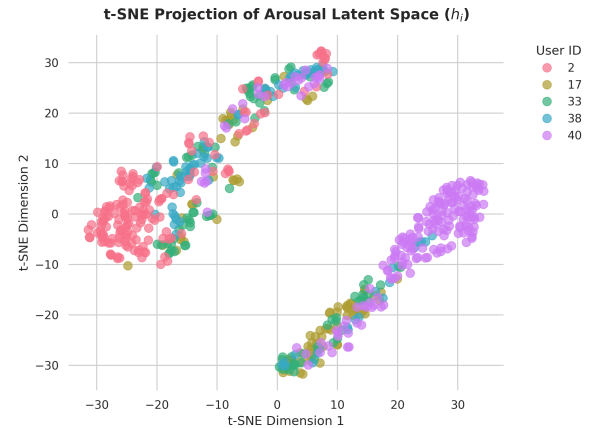


Figure 2: **t-SNE Visualization.** A two-dimensional projection of the Arousal stream’s latent space (h_i). The high degree of overlap among different user clusters visually confirms that the DANN successfully forced the network to learn user-agnostic affective representations.

To visually confirm this adversarial effect, we present a t-SNE projection of the Arousal stream’s latent representations (Figure 2). The plot illustrates a highly overlapping, user-agnostic feature space, confirming that the Gradient Reversal Layer successfully stripped stylistic identity markers from the embeddings.

5.2 Modality and Ablation Analysis

Isolating input modalities exposed a distinct bias in feature utilization. For Valence, the model maintained consistent performance regardless of modality ($r_c = 0.655$ for Words, 0.627 for Essays). Conversely, Arousal prediction exhibited a heavy dependency on explicit emotional lexicon, scoring $r_c = 0.572$ on Words but collapsing to 0.307 on Essays. This suggests that while the backbone extracted contextual hedonic tone from narratives, it struggled to infer emotional intensity without explicit anchors.

Internal ablation studies further confirmed the necessity of our preprocessing protocols. Bypassing the Out-of-Fold (OOF) sanitation degraded overall performance, validating that the inverted labels corrupted early representation learning.

6 Conclusion

In this paper, we presented the LexMachina framework for longitudinal affect assessment in SemEval-2026 Task 2. Our dual-stream architecture, anchored by a disentangled DeBERTa-v3 backbone, successfully navigated the complexities of predicting dynamic emotional shifts from ecological text. By implementing a rigorous Out-of-Fold (OOF) data sanitation protocol, we effectively insulated our representation learning from severe label noise and semantic inversions present in the training corpus. Furthermore, our bifurcated optimization strategy yielded highly competitive composite correlations. Most notably, our system demonstrated exceptional zero-shot generalization capabilities on the Arousal dimension. By employing adversarial disentanglement, we proved that it is possible to strip away stylistic user biases and successfully extract universal markers of emotional intensity from entirely unseen populations.

7 Future Work

To address the current bottlenecks in our pipeline, our future work will focus on architectural refinements for both the Valence and Arousal streams.

For Valence, we propose replacing the post-hoc isotonic calibration with a dedicated, custom **Attention Head** positioned immediately after the DeBERTa backbone. This supplementary attention mechanism would be explicitly optimized

to attend to high-polarity affective cues, dynamically amplifying the latent representations. This would allow the model to naturally "stretch" its predictions toward the $[-2, 2]$ extremes during the forward pass, effectively solving the conservative bias without relying on artifact-inducing, non-parametric calibration functions.

For Arousal, we intend to transition from a static DANN to an **Instance-Weighted Domain Adaptation** strategy. We propose augmenting the Gradient Reversal Layer (GRL) with a dynamic penalty function parameterized by two historical metrics: (1) the volume of a user's past texts, and (2) their affective "reliability" (an inverse measure of their intra-individual variability). By mathematically tying the adversarial penalty (λ) to this reliability score, unseen or highly erratic users would trigger strict adversarial disentanglement, while stable, known users would bypass the GRL. This dynamic gating would allow the network to seamlessly interpolate between zero-shot generalization for strangers and personalized fine-tuning for known subjects.

8 Limitations

While our framework exhibits strong performance, it introduces two critical architectural trade-offs.

First, the strict decoupling of user identity enforced by the DANN in the Arousal stream imposes a severe penalty on *seen* subjects. By actively unlearning stylistic idiosyncrasies to accommodate the 60.2% unseen test volume, the model discards the highly predictive historical baselines of known individuals. This results in a pronounced performance disparity, where unseen users significantly outperform known users on the intensity axis.

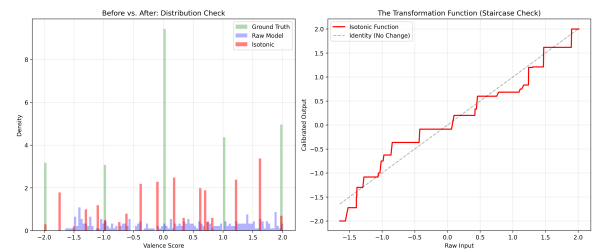


Figure 3: **Valence Forensic Analysis.** Visualization of the quantization artifacts (“staircase effect”) introduced by the Post-Hoc Isotonic Calibration.

Second, our reliance on Post-Hoc Isotonic Calibration to correct the Valence regression head's "regression to the mean" introduced detrimental quantization artifacts (illustrated in Figure 3). Because isotonic regression maps continuous outputs to a discrete step function based on the validation set's cumulative distribution, it artificially discretized the continuous output space. This manifested as a visible "staircase" effect in our error analysis, occasionally triggering severe error spikes when predicting borderline or highly nuanced emotional states that fell between the learned calibration steps.

References

- Gonalo Azevedo Mendes and Bruno Martins. 2023. Quantifying valence and arousal in text with multilingual pre-trained transformers. *Advances in Information Retrieval (ECIR)*, pages 243–257.
- Yaroslav Ganin, Evgeniya Ustinova, Hana Ajakan, Pascal Germain, Hugo Larochelle, Franois Laviolette, Mario Marchand, and Victor Lempitsky. 2016. Domain-adversarial training of neural networks. *The Journal of Machine Learning Research*, 17(1):2096–2130.
- Pengcheng He, Xiaodong Liu, Jianfeng Gao, and Weizhu Chen. 2020. DeBERTa: Decoding-enhanced BERT with Disentangled Attention. *arXiv preprint arXiv:2006.03654*.
- Yi-Min Jian, et al. 2025. CYUT-NLP at ROCLING-2025 Shared Task: Valence-Arousal Prediction in Physicians' Texts Using BERT, RAG, and Multi-Teacher Pseudo-Labeling. *Proceedings of ROCLING*, pages 321–328.
- Michail Mitsios, et al. 2024. Improved Text Emotion Prediction Using Combined Valence and Arousal Ordinal Classification. *Proceedings of NAACL*, pages 512–519.
- Melissa Moreno Novoa, et al. 2025. UTBNLP at SemEval-2025 Task 11: Predicting Emotion Intensity with BERT and VAD-Informed Attention. *Proceedings of SemEval*.
- James A. Russell. 1980. A circumplex model of affect. *Journal of Personality and Social Psychology*, 39(6):1161.

Bianca Zadrozny and Charles Elkan. 2002. Transforming classifier scores into accurate multiclass probability estimates. *Proceedings of SIGKDD*.